

Nguyen Minh Khoa

AI Engineer & Researcher

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I'm Nguyen Minh Khoa, an AI engineer and Research Assistant at AiTA Lab, FPT University.

My recent work includes handwritten mathematical expression recognition and text-to-molecule generation. I work across data preparation, model experiments, and inference integration, with a preference for research ideas that can be tested in working software.

My current focus is AI for intelligent manufacturing. I look for work that gives equal weight to careful experiments and hands-on implementation.

I'm open to AI engineering, research engineering, and applied machine-learning collaborations.

RESEARCH EXPERIENCE

Research Assistant | AiTA Lab, FPT University | Oct 2024 - Present

- First-authored LexiChem, accepted at APWeb-WAIM 2026, and presented it in Da Nang. Developed method ideas, ran experiments and ablations, wrote the Introduction and Results & Discussion, and co-wrote Methods.
- Co-authored Mask CoMER (ICDAR 2025) and co-authored the MTL + CBARM journal paper, contributing pretraining and multi-task learning experiments for mathematical expression recognition.
- Co-first-authored ChemAligner-T5 (ICCIES 2026), a study aligning text and molecular representations for text-to-molecule generation.
- Reviewed and refined the methodology for CorrTie, an active-learning paper published in the APWeb-WAIM 2026 proceedings, as a co-author.

RESEARCH & ENGINEERING PROJECTS

LexiChem | AI & Inference Contributor | 2025 - 2026

Completed capstone | Three-person team | Local application

- Built the AI path in a three-person capstone, bringing text-to-molecule models into an application where users generate, compare and inspect molecular candidates in 2D/3D.
- Created training-ready SELFIES data from Mol-Instructions, L+M-24 and ChEBI-20 by validating structures, canonicalizing SMILES and removing duplicates with RDKit.
- Developed shared-latent text/SELFIES alignment ideas and tested them through experiments and ablations, forming the research core of the first-author APWeb-WAIM paper.
- Integrated FastAPI/Triton inference, RDKit structure checks and MongoDB storage into Experiments, connecting prompt entry to validated molecules and saved runs.

Tools: Python, PyTorch, Transformers, RDKit, SMILES/SELFIES, FastAPI, NVIDIA Triton, MongoDB

NexOps | AI, Simulation & Platform Lead | 2025 - Present

Working investment/demo prototype | Local MES/SCADA runtime

- Built the local runtime linking machine telemetry, work orders, production plans, alarms and OEE across operator views, leading the project's AI, simulation and platform work.
- Created repeatable machine and shift simulations to exercise planning and alarm workflows under the same factory scenarios across runs.
- Built application-side MQTT/EMQX integration through FastAPI and TimescaleDB, coordinating with the IoT teammate to connect simulator and edge events to live dashboards.
- Integrated a Bambu Lab printer over LAN and demonstrated device control in a recorded trial, extending the prototype beyond simulated machines.

Tools: Python, FastAPI, MQTT/EMQX, PostgreSQL/TimescaleDB, React, TypeScript, Docker Compose

RESEARCH & ENGINEERING PROJECTS - CONTINUED

HMER | Research Contributor | 2024 - 2026

Handwritten mathematical expressions to LaTeX | *AiTA Lab*

- Implemented Mask CoMER methods, reproduced baselines and contributed experiment design, ablations, error analysis and paper writing as a co-author.
- Co-authored the MTL + CBARM paper in the Journal of Information and Telecommunication through multi-task learning experiments combining structural supervision and attention refinement.

Tools: Python, PyTorch, PyTorch Lightning, CUDA, Weights & Biases, CROHME

PUBLICATIONS

Published as **Khoa Minh Nguyen**.

LexiChem: Shared-Latent Alignment for Faithful Text-to-Molecule Generation in SELFIES Space

APWeb-WAIM 2026 | First author | Accepted & presented

Developed method ideas and ablations; wrote the Introduction and Results & Discussion, and co-wrote Methods. Presented at APWeb-WAIM in Da Nang.

Mask CoMER: Enhancing Handwritten Mathematical Expression Recognition with Masked Language Pretraining and Regularization

ICDAR 2025 | Co-author | Published

Method implementation, baseline reproduction, ablations, error analysis and paper writing. Up to +5 ExpRate points over CoMER on CROHME.

ChemAligner-T5: A Unified Text-to-Molecule Model via Representation Alignment

ICCIES 2026 | Co-first author | Published

Co-first-author work on aligning text and molecular representations with BioT5+, evaluated on L+M-24. Paper reports BLEU 69.77% and Levenshtein distance 31.28% on L+M-24.

Enhancing handwritten mathematical expression recognition with convolutional block attention refinement module and multi-task learning

Journal of Information and Telecommunication | Co-author | Published online 19 August 2026

Experimental contributions to multi-task learning for HMER with structural supervision and attention refinement.

CorrTie: Correction-aware Tie-breaking for Active Learning with Vision-Language Models

APWeb-WAIM 2026 | Co-author | Published

Critical review and methodological refinement of correction-aware sample selection for vision-language active learning.

EDUCATION

Bachelor of Science in Artificial Intelligence | FPT University, HCMC

2022 - 2026 | Academic requirements completed; degree conferral expected November 2026.

AWARDS & SCHOLARSHIPS

50% Tuition Scholarship | FPT University | 2022

Earned through the university scholarship examination, covering half of tuition for the full programme.

Second Prize | RESFES HCMC | 2026

Awarded to the ChemAligner-T5 team at FPT University HCMC's Spring 2026 student research competition.

Second Prize | RESFES HCMC | 2025

Awarded to the Mask CoMER team at FPT University HCMC's student research competition.

VND 50 Million Team Award & Funding | Demo Pitching Day | 2025

TrendRadar team recognition and project funding from FPT University.

VND 40 Million Team Support | Innovation Quest | 2025

TrendRadar, FrontAI team, through SIHUB; includes a VND 10 million team reward.

Silver Medal | HCMC Open April Olympiad | 2021

Mathematics, Grade 11.

TECHNICAL SKILLS

AI & research: Python, PyTorch, Transformers, model experiments and ablations, HMER, cross-modal learning, text-to-molecule generation

Systems & data: FastAPI, NVIDIA Triton, RDKit, PostgreSQL/TimescaleDB, MongoDB, MQTT, Docker, Git

CERTIFICATIONS

AIO | AI VIET NAM: Machine Learning; Deep Learning; Computer Vision & NLP; GenAI & LLMs.

Machine Learning | DeepLearning.AI · Stanford Online | 2023

Deep Learning | DeepLearning.AI | 2024

Natural Language Processing | DeepLearning.AI | 2025

IBM Full Stack Software Developer | IBM | 2024

Additional AI training: DeepLearning.AI TensorFlow Developer; Big Data; Ethics in the Age of AI; AI Foundations for Everyone; Introduction to AI.

Software & cloud: AWS Cloud Technical Essentials; Software Development Lifecycle.

Research & academic writing: Research Methodologies; Being a researcher (in Information Science and Technology); Understanding Research Methods; Introduction to Research for Essay Writing; Advanced Writing; Academic Skills for University Success.

Project management: Project Management Principles and Practices.